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Project Title:	Identification of the changes in the brain proteome associated with HZE-induced neurocognitive impairment		
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Flight Assignment:	NOTE: Extended to 12/31/2014 per PI (Ed., 1/23/14)		
Key Personnel Changes/Previous PI:	November 2012 report: Dr. Sucharita Dutta, a proteomics expert, has been added to the team.		
COI Name (Institution):	Semmes, Oliver (Eastern Virginia Medical School) Dutta, Sucharita (Eastern Virginia Medical School)		
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	One of NASA's biggest concerns about the long-term health of astronauts who go on deep space missions is the impact that Hze (high energy particle) radiation will have on brain function (neurocognition). Data from our laboratory and others suggests that there is significant impairment of certain neurocognitive tasks following exposure to low Hze doses. The goal of this application is to use proteomic profiling/imaging to establish the mechanistic basis of Hze-induced cognitive impairment. Our underlying hypothesis is that Hze-induced spatial memory impairments (HISMI) or Attentional Set Shifting Impairments (HIASSI) are due to Hze-induced changes in the proteome of the brain (neuroproteome). Our sub-hypothesis is that the use of a novel technique, called MALDI-MSI, will identify specific regions of the brain where the neuroproteome differs in rats that have developed HISMI and HIASSI. We further hypothesize that proteomic profiling of serum and cerebrospinal fluid (CSF) will identify proteins that can be used as		

Task Description:	biomarkers to monitor the emergence of HISMI and HIASSI. We also hypothesize that different mechanisms of HISMI and HIASSI occur following exposure to Hze radiations with different track structures. Our studies will thus address Central Nervous System (CNS) Gap 2, and we shall specifically focus of the following aims: Aim 1. Identify changes in the neuroproteome that are associated with HISMI and HIASSI. Aim 2. Identify proteins that are present in the serum and CSF that are biomarkers of an individual rat¿s susceptibility to HISMI/HIASSI. Aim 3. Determine the linear energy transfer (LET)-dependency for HISMI and HIASSI, and whether the threshold dose is modified by prior exposure to protons, and if the underlying mechanisms of HISMI and HIASSI are constant irrespective of the LET of the Hze particle. In Aim 1, two groups of rats (5-week and 9 month old males) will be irradiated with 20 cGy Hze (1 GeV 56Fe) irradiation, and HISMI and HIASSI assessed at 3 months post irradiation. The rats will then be classified as normal or as having HISMI or HIASSI, then sacrificed, and their brains recovered. We shall use MALDI-IMS to analyze changes in the neuroproteome at the regional level. Subsequently, we shall identify specific proteins whose expression changes when HISMI or HIASSI have emerged, and will use immunohistochemcial analysis to determine the specific brain tissue (e.g., the dentate gyrus or CA2) affected. These studies will give considerable insight into the underlying cause for Hze-induced neurocognitive failure. In Aim 2, biological fluids (CSF and serum) from the rats used in Aim 1 will be recovered, and then subjected to MALDI-TOF/TOF proteomic profiling to identify unique proteins that are prognostic markers of an individual's rat's susceptibility to develop HIMSI and HIASSI. These proteins could be potentially used for post-mission surveillance of astronauts to monitor the emergence of Hze-induced neurocognitive impairments. In Aim 3, we shall repeat the studies outlined in Aim 1, but determine the impact that exposure to other Hze particles (300 MeV 37Cl, 1 Gev 48Ti , 400 MeV 48Ti) have on the induction of HISMI and HIASSI, and determine whether the mechanism of HIASSI and HISMI is the same following exposure to these Hze radiations as was observed in Aim 1. These studies will be confined to the aged rat population, as these resemble the biological age of the current astronaut cadre. We shall also determine if prior irradiation with 1 Gy of protons modifies the threshold dose for HISMI and HIASSI and the spectrum of proteomic changes that occur when HIASSI and HISMI emerge. Due to logistical constraints these combined proton/Hze exposure studies will be confined to 400 MeV 48Ti and 1 GeV 56Fe. As part of the no-cost extension we determined if the introduction of an exercise regimen would have any impact on the severity of documented HZE-induced impairment of spatial memory. Collectively, these studies will provide considerable insight into the underlying mechanism of Hze-induced neurocognitive failure and potentially provide a post mission screen for HISMI and HIASSI in astronauts returning from deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	Current models predict that the astronauts on deep space missions will be exposed to ~25 cGy of Galactic Cosmic Radiation (GCR). The long-term consequence of being exposed to such GCR doses is largely unknown, but recent data from our laboratory and that of Dr. Rabin, suggest that low (<20 cGy) doses of HZE particles lead to significant impairments of various neurocognitive functions. We have shown that exposure of young rats to 20 cGy of 1 GeV/n 56Fe leads to a significant impairment in spatial memory (Britten et al., 2012) and attentional set shifting (ATSET) performance (Lonart et al., 2012). Data from Dr. Rabin suggest that the LET dependency for neurocognitive impairment may differ substantially from that seen for cell killing. We thus determined the LET-dependency for impairment of spatial memory performance and ATSET in rats that have a comparable biological age to the Astronaut population. Our studies had to be confined to just male rats due to logistical constraints and the amount of work required to minimize the variable estrogen levels in female rats. We chose to use Retired Breeder rats for these studies. Such rats typically have a chronological age equivalent to a 30 year old human, but older rats tend to be obese, and can be lethargic due to their prolonged time in captivity. Thus while retired breeder rats may have a chronological age lower than that of many Astronauts, we believe that they are more representative of the physical and mental fitness status of Astronauts. Data was obtained on the performance of rats in the Barnes' Maze (spatial memory) and the ATSET test following exposure to mission-relevant (5-20 cGy) of 1 GeV/n 56Fe, 1 GeV/n 48Ti, and 600 MeV/n 28Si (respective LETs: 150, 106, and 51 keV/ µm). Data was also obtained on the changes in the composition of the neuroproteome associated with the HZE-induced impairment of these neurocognitive processes. Our efforts primarily focused on determining the changes in the proteome of the Hippocampus in rats that have good or poor spatial memory performance after exposure to 20 cGy of 1 GeV/n 56Fe and 1 GeV/n 48Ti particles. In some instances, the impact of HZE exposure on the Readily Releasable Pool of neurotransmitters (GABA, Choline, and Glutamate) in the Hippocampus and Basal Forebrain was also established. MATERIALS & METHODS. Retired breeder (6-9 months old) male Wistar rats were exposed to 5, 10, 15, or 20 cGy of 1 GeV/n 56Fe, 1 GeV/n 48Ti, or 600 MeV/n 28Si particles at Brookhaven National Laboratory (BNL). At 3 months post exposure the ability of rats to conduct ATSET and spatial memory was established. After completion of the behavioral tasks the animals were sacrificed and the brains recovered for subsequent neurotransmitter or proteomic analysis on the Q-Exactive platform. RESULTS. Spatial memory impairment. All of the HZE species studied to date, with LETs ranging from ~50-150 KeV/µm, resulted in impairment of Spatial memory performance at mission-pertinent doses. Exposure to 10 cGy of all three ion species resulted in significant impairment of spatial memory. Exposure to 5 cGy of 1 GeV/u 56Fe had no measurable impact on spatial memory performance; we have insufficient data to indicate whether exposure to 5 cGy of the other ions resulted in spatial memory impairment. A marked inter-individual susceptibility to develop spatial memory impairment was observed following exposure to all ion species. This dichotomy in Spatial memory impairment raises questions about the validity of population based averages to identify "threshold" doses for spatial memory impairment, and we propose that

Task Progress:	in future the use of “individuals at risk” may be a more meaningful means for assessing the risk of spatial memory impairment. Exposure to 10 cGy of GeV/n 56Fe or 1 GeV/n 48Ti resulted in 15-25% of rats exhibiting significant (Z score >2) impairment of spatial memory. There is no obvious variation in the potency of these HZE particles to induce spatial memory impairment. Attentional Set shifting impairment. We recently published (Lonart et al., 2012) data from young rats that had received cranial irradiation with 1 GeV/μ 56Fe, where we reported that exposure to 20 cGy 1 GeV/μ 56Fe resulted in the majority of rats being incapable of completing all stages of the ATSET test, with many rats failing at the Compound Discrimination Reversal (CDR) stage. Whole body exposures of young rats resulted in a similar pattern of failures. Sixty percent of the sham treated rats completed all stages of testing whereas only 20% of the irradiated rats could complete all stages of the task. However, these rats have a biological age comparable to that of teenagers and our primary mandate was to determine if rats with a comparable age to the Astronauts were equally sensitive to the effects of these low doses of 1 GeV/n 56Fe, 1 GeV/n 48Ti, and 600 MeV/n 28Si particles. The studies with the Aged rats were complicated by a marked decline in the ability, or the motivation of sham-irradiated Aged rats to conduct Simple Discrimination (SD: regulated by the medial prefrontal cortex) and CDR (regulated by the basal forebrain). Exposure to 15 cGy of 1 GeV/n 56Fe resulted in multiple deficits in ATSET performance above that seen in the sham-irradiated rats. Should similar defects be induced in Astronauts this would result in: • SD impairment- a decreased ability to multi-task successfully; • Compound Discrimination (CD) impairment- a decreased ability to identify and focus on relevant aspects of the task being conducted; • CDR Impairment- an inability to relearning (“reverse”) an attentional set once it is formed. Such mental inflexibility would result in an individual “sticking to their guns” about what parameter is important to the successful completion of a task, even when that parameter has switched from a positive factor to a negative factor. We have shown that there is a reduction in the Choline Readily Releasable pool within the Basal Forebrain, which may be responsible for the impairment of CDR performance by 1 GeV/n 56Fe (Britten et al., 2014). Exposure to 1 GeV/n 48Ti and 600 MeV/n 28Si resulted in impairments in ATSET performance at lower does than was observed with 1 GeV/n 56Fe. Rats exposed to 10 cGy 1 GeV/n 48Ti had significantly impaired CD performance, and Intra-dimensional set shifting (IDS) (Britten et al., 2014). The functional consequence of the impairment of IDS performance would be a reduction in the individual’s cognitive flexibility (the mental ability to adjust thinking or attention in response to changing goals and/or environmental stimuli). Rats exposed to 10 cGy 600 MeV/n 28Si had significantly impaired performance in the SD, CD, CDR, and Intra-dimensional set shifting reversal (IDR) paradigms. These data provide further support for the notion that lower LET particles (1 GeV/n 48Ti and 600 MeV/n 28Si) are more potent at inducing neurocognitive impairments than 1 GeV/n 56Fe particles, and extend the findings of Rabin et al., 2007, to encompass neurocognitive tasks regulated by the cortex as well as the Striatum and Amygdala. Hippocampal proteomic changes. For this work we partnered with The Biomarkers Research Initiatives in Mass Spectrometry (BRIMS) Center, a Thermo Fisher Scientific (http://www.thermoscientific.com) center of excellence that develops and promotes comprehensive, integrated, and robust mass spectrometry (MS)-based workflows that links early-stage discovery to next-stage quantitative verification of protein, peptide, and small molecule biomarkers. Using this workflow we have been able to obtain quantitative data on ~900 proteins in most hippocampal samples studied. This wealth of data has allows us to determine some of the changes in the hippocampal proteome of the rats exposed to mission relevant doses of 1 GeV/μ 56Fe particles, and the specific changes observed in 20/Impaired and 20/Functional rats has provided some insight into the functionality of multiple pathways in these rats. While quantitative data has been derived on ~1000 proteins, it must be remembered that the proteomic profiling technique used in this study, while being one of the most sensitive currently available, has not detected low abundance proteins that are likely to be the key rate limiting steps in many pathways. Nonetheless, there is clear evidence that hippocampi exposed to the HZE radiation have persistent oxidative stress, altered adenosine metabolism, and may have on-going GluR1-mediated neuronal death. However, the ultimate phenotypic outcome (functional vs. impaired spatial memory performance) appears to be determined by either the up-regulation of pathways that serve to ameliorate the aforementioned deficits, by altering brain plasticity (20/Functional rats), or failure to invoke, or even lose key proteins in those pathways (20/Impaired). This work has been submitted for publication. SUMMARY. Collectively, these data suggest that mission-relevant doses (10-15 cGy) of HZE particles with LETs of 51-150 KeV/μm induce significant neurocognitive defects (spatial memory performance and ATSET) in rats of comparable biological age to the astronaut population. We have shown that HZE exposure alters the neurotransmitter release properties of the basal forebrain and have identified specific changes in the hippocampal proteome that are associated with either good or bad spatial memory performance in HZE-irradiated rats, which may provide some insight into the mechanisms that govern individual susceptibility to HZE-induced cognitive impairment.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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